

Laptop Audio Amplifier

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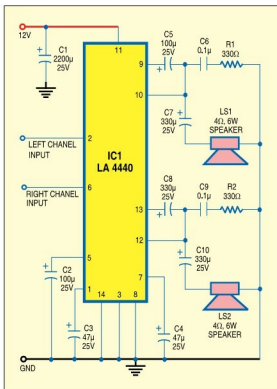
Usually, the audio output from a laptop's built-in speakers is low. A power amplifier is required to get a high volume. Here is a simple circuit to amplify the laptop's audio output.

The circuit is built around power amplifier IC LA 4440 (IC1) and a few other components. LA4440 is a dual-channel audio power amplifier. It has low distortion over a wide range of low to high frequencies with good channel separation. Inbuilt dual channels enable it for stereo and bridge amplifier applications.

In dual mode LA4440 gives 6 watts per channel and in bridge mode 19-watt output. It has ripple rejection of 46 dB. The audio effect can be realised

by using two 6-watt speakers. Connect pins 2, 6 and ground of IC1 to the stereo jack which is to be used with the laptop.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. The circuit works off regulated 12V power supply. It is recommended to use audio input socket in the circuit board. Use a proper heat-sink for LA4440. ●



SMPS Spark Suppressor

RAJU BADDI

This circuit protects the power plug of a switched-mode power supply (SMPS) against sparks that may damage its contact terminals. It normally happens when the device

(laptop charger, mobile charger, etc) is plugged to a multi-socket board whose power is already turned on. Under this condition, while plugging in the SMPS, the capacitor (which is present in the initial stage of the power input circuitry) inside it conducts heavily for

a short time, virtually short-circuiting power terminals L and N.

The time of short circuit depends upon the time constant of the circuit formed by the input capacitor and resistor. It can cause a heavy spark

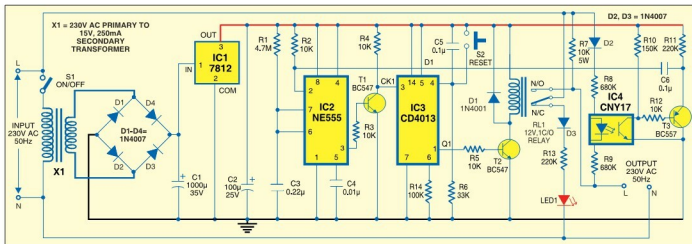


Fig. 1: Circuit of SMPS spark suppressor